MIL-F-63446A (AR)
12 September 1983
SUPERSEDING
MIL-F-63446(AR)
9 August 1982

MILITARY SPECIFICATION

FUZES, GENERAL PURPOSE GRENADE: M230 AND M231 (PRACTICE)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification contains requirements and quality assurance provisions for parts, assembly and packing of the M230 and M231 fuzes with detonator for the M73 and M75 General Purpose Grenades, respectively.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-591	Aluminium Alloy Die Casting

MILITARY

MIL-D-14978	Detonator, Stab, M55; Loading Assembling and Packing.
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-810	Environmmental Test Methods
MIL-STD-1167	Ammunition Data Cards

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

Code of Federal Regulations

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402; orders shall specify 49 CFR 100-199.)

* DRAWINGS

U.S. ARMY RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

8799112	Marking Diagram and Sealing for Interplant or CONUS Shipment of Ammunition Fiber Board Boxes
9333825	Fuze Assembly - M230 All ways.
9340491	Box, Fiber, Inner, Packing, Ammunition, for Fuze, M230/M231
9340492	Box, Fiber, Outer, Packing, Ammunition for Fuze, M230/M231
9345168	Fuze Assembly - M231 All Ways

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the fuze. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative. The dimensions shown on the drawings apply at temperatures from 68°F to 70°F.

3.2 First Article. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Materials. Materials shall be as specified herein and on the applicable drawings.

* 3.4 Fuze assemblies and parts. fuze assemblies and parts shall comply with all requirements specified on Drawings 9333825 and 9345168 and associated drawings as applicable and with all requirements specified in applicable specifications and standards.

3.4.1 Molded and plastic parts. Molded and plastic parts (slider) made from each cavity of the mold shall initially be inspected with increased frequency until it has been established that each cavity is capable of producing sufficiently accurate parts. In verifying the capability of the mold, prior to producing any parts in quantity, 500 plastic sliders per cavity shall be produced, kept separate and appropriately labeled. From

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

each group of sliders, 50 parts shall be randomly selected and checked dimensionally for compliance to the drawings and specification. Two discrepant parts shall reject the preproduction lot. Production shall not commence until a preproduction lot is accepted. Upon acceptance of the preproduction lot, subsequent lots may be inspected at a reduced frequency. Unless otherwise specified in the contract, at least 0.1 percent of the molded parts subsequently made shall be measured and gaged. A single reject shall reject the lot.

3.4.2 Die cast parts. The aluminum die cast fuze body shall be manufactured, tested and inspected per the requirments of Federal Specification QQ-A-591 and Drawing 9333824.

3.4.3 Porosity check. Each aluminum die cast fuze body of each production lot shall be inspected for porosity in the firing pin bore.

3.4.4 Springs.

3.4.4.1 Spring embrittlement. The springs used in the fabrication of the fuze shall not break when extended or distorted to the point of minimum permanent distortion.

3.4.4.2 Spring set. The slider spring and firing pin spring shall return to within .015 inch of the original free length when compressed at elevated temperatures.

3.4.5 Detonator. All detonators shall have been subjected to and passed all the requirements of MIL-D-14978.

3.4.6 Lead output. The long and short lead assemblies, when initiated in tandem by an M55 detonator, shall cause an impression in a steel witness block not less than .125 inch in diameter and .015 inch in depth.

3.5 Functional requirements.

3.5.1 Jolt. The fuze shall be safe to store, handle, use or dispose of after being subjected to Test 101.2, Procedure 1 of MIL-STD-331.

3.5.2 Jumble. The fuze shall be safe to store, handle, use or dispose of after being subjected to Test 102.1 of MIL-STD-331.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.5.3 Transportation vibration. The fuze shall be safe and meet all the functional requirements of this specification after being subjected to Procedure 1 Test 119 of MIL-STD-331.

3.5.4 Helicopter vibration. The fuze shall be safe and meet all the functional requirements of this specification after being subjected to Method 514 of MIL-STD-810.

3.5.5 Five foot drop. The fuze shall be safe to store, handle, use or dispose of after being subjected to five foot drop in accordance with Test 111.1, Procedure 2 of MIL-STD-331. The fuze need not be operable.

3.5.6 Thermal shock. The fuze shall be safe and meet all the functional requirements of this specification after being subjected to Test 113.1 of MIL-STD-331.

3.5.7 Shear wire test. It shall require 18 ± 1.5 pounds of force to sever the shear wire of an assembled fuze when a pull load is applied in a direction within 2 degrees parallel to the axis of the arming pin.

3.5.8 Safety pin removal. It shall require not less than one pound and not more than two pounds of force to remove the safety pin from the fuze when the direction of the applied load is within 5 degrees of the axis of the safety pin.

3.5.9 Arming time. After removal of the Safety Pin and upon withdrawal of the Arming Pin, the slider shall travel the full arming stroke in an elapsed time of 1.75 ± .25 seconds.

3.5.10 Fuzing - No Go. An armed fuze shall not trigger (release firing pin) when subjected to the No Go Shock Test.

3.5.11 Fuzing - Go. An armed fuze shall trigger and initiate the detonator when subjected to the Go Shock Test.

3.5.12 Reliability. The fuze shall demonstrate a minimum acceptable value of 94.6% with a best operational capability of 99% at 90% confidence.

3.5.13 Temperature. The fuze shall withstand exposure to and function after a storage temperature of -50°F ± 5°F, 70°F ± 15°F and 160°F ± 5°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.6 Data requirements. The contractor shall generate data in accordance with the requirements of 4.5.17.

3.7 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are made a part of this specification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

* 4.3.1 Submission. the contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of the following items in sample quantities as indicated:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

PART DESCRIPTION	DRAWING	QUANTITY
Slider	9333807	50/Cavity
Body, Fuze	9333824	50/Cavity
Spring, Slider	9333811	50
Screw, Pivot	9333808	25
Fuze Subassembly No. 1	9347183	25
Anti-Twist Pin	9347220	25
Fuze Subassembly No. 2	9347184	25
Sleeve, Fuze Body	9345131	25
Fuze Subassembly No. 3	9347185	25
Fuze Subassembly No. 4	9347186	50
Trigger	9333812	25
Pin, Arming	9333814	25
Wire, Shear	9345128	25
Slider, M55 Detonator Assembly	9347219	25
Pin, Safety	9333818	25
Pinion, Escapement	8833366	25
Gear, Escapement	8833365	25
Escapement Gear-Pinion Assembly	8833364	25
Pinion No. 1	8833360	25
Gear No. 1	8833359	25
No. 1 Gear-Pinion Assembly	8833358	25
Pinion No. 2	8833363	25
Gear No.2	8833362	25
No. 2 Gear Pinion Assembly	8833361	25
Lever	8833369	25
Pallet, Lever	8833368	25
Shaft, Lever	8833370	25
Lever Assembly	8833367	25
Plate, Bottom	8833356	25/die
Plate, Top	8833357	25/die
Pillar	8833355	25
Escapement Gear Assembly	8833353	50
Fuze Subassembly No. 5	9347187	25
Body, Firing Pin	9333816	25
Pin, Straight	9345321	25
Firing Pin Complete	9345130	50
Cap, Firing Pin	9333817	25
Spring, Firing Pin	9333810	50
Firing Pin Assembly	9347182	25
Ball, Detent	9349223	25
Fuze Subassembly No. 6	9347188	25
Plate, Barrier	9333809	25
Spring, Arming Pin Lock	9345129	25
Disc. Closing	9333821	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

PART DESCRIPTION	DRAWING	QUANTITY
Lead Cup, Short	9333822	50
Lead Cup Assembly, Short	9333823	50
Lead Cup, Long	9333819	25
Lead Cup Assembly, Long	9333820	50
Plate Assembly, Barrier	9345166	25
Fuze Assembly, M231 Allways	9345168	239
Fuze Assembly, M230 Allways	9333825	239

* 4.3.2 Inspections to be performed. See MIL-A-48078 and table I specified herein.

4.3.3. Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 11		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Slider (Dwg. 9333807)				
	Examination for defects	50/ Cavity		3.4.1	4.4.2.1
	Body, Fuze (Dwg. 9333824)				
	Examination for defects	50/ Cavity		3.4.2	4.4.2.2
	Porosity in firing pin Bore	50/ Cavity		3.4.3	4.5.1
	Spring, Slider (Dwg. 9333811)				
	Examination for Defects	50		3.4	4.4.2.3
	Spring Load Requirement	25		3.4	Gage
	Spring Embrittlement	25		3.4.4.1	4.5.2
	Spring Set	25		3.4.4.2	4.5.3
	Screw, Pivot (Dwg. 9333808)				
	Examination for Defects	25		3.4	4.4.2.4

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuzes, General Purpose Grenade: M230 and M231		SHEET 2 OF 11		Listed below
	Fuze Subassembly No. 1 (Dwg. 9347183)				
	Examination for Defects	25		3.4	4.4.2.5
	Torque requirement	25		3.4	Gage
	Anti-Twist Pin (Dwg. 9347220)				
	Examination for Defects	25		3.4	4.4.2.6
	Fuze Subassembly No. 2 (Dwg. 9347184)				
	Examination for Defects	25		3.4	4.4.2.7
	Minimum pushout force	25		3.4	Gage
	Sleeve, Fuze Body (Dwg. 9345131)				
	Examination for Defects	25		3.4	4.4.2.8
	Fuze Subassembly No. 3 (Dwg. 9347185)				
	Examination for Defects	25		3.4	4.4.2.9
	Pushout requirement	25		3.4	Gage

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 3 OF 11		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze Subassembly No. 4 (Dwg. 9347186) Examination for Defects	50		3.4	4.4.2.10
	Trigger (Dwg. 9333812) Examination for Defects)	25		3.4	4.4.2.11
	Pin, Arming (Dwg. 9333814) Examination for Defects	25		3.4	4.4.2.12

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 4 OF 11		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Wire, Shear				
	(Dwg. 9345128)				
	Examination for Defects	25		3.4	4.4.2.13
	Shear load	25		3.4	Gage
	Slider, M55 Detonator Assembly				
	(Dwg. 9347219)				
	Examination for Defects	25		3.4	4.4.2.14
	Detonator Deterioration	25		3.4.5	4.5.4
	Pin, Safety				
	(Dwg. 9333818)				
	Examination for Defects	25		3.4	4.4.2.15
	Pinion, Escapement				
	(Dwg. 8833366)				
	Examination for Defects	25		3.4	4.4.2.16
	Gear Escapement				
	(Dwg. 8833365)				
	Examination for Defects	25		3.4	4.4.2.17

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231	5 of 11	Listed below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Escapement Gear-Pinion Assembly (Dwg. 8833364) Examination for Defects	25		3.4	4.4.2.18
	Pinion No. 1 (Dwg. 8833360) Examination for Defects	25		3.4	4.4.2.19
	Gear No. 1 (Dwg. 8833359) Examination for Defects	25		3.4	4.4.2.20
	No. 1 Gear-Pinion Assembly (Dwg. 8833358) Examination for Defects	25		3.4	4.4.2.21
	Pinion No. 2 (Dwg. 8833363) Examination for Defects	25		3.4	4.4.2.22
	Gear No. 2 (Dwg. 8833362) Examination for Defects	25		3.4	4.4.2.23

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 6 OF 11		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	No. 2 Gear-Pinion Assembly (Dwg. 8833361) Examination for Defects	25		3.4	4.4.2.24
	Lever (Dwg. 8833369) Examination for Defects	25		3.4	4.4.2.25
	Pallet, Lever (Dwg. 8833368) Examination for Defects	25		3.4	4.4.2.26
	Shaft, Lever (Dwg. 8833370) Examination for Defects	25		3.4	4.4.2.27
	Lever Assembly (Dwg. 8833367 Examination for Defects	25		3.4	4.4.2.28
	Plate, Bottom (Dwg. 8833356) Examination for Defects	25/die		3.4	4.4.2.29
	Plate, Top (Dwg. 8833357) Examination for Defects	25/die		3.4	4.4.2.30

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 7 OF 11		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pillar (Dwg. 8833355) Examination for Defects	25		3.4	4.4.2.31
	Escapement Gear Assembly (Dwg. 8833353) Examination for Defects	50		3.4	4.4.2.32
	Axial Pull of Assembly	50		3.4	Gage
	Fuze Subassembly No. 5 (Dwg. 9347187) Examination for Defects	25		3.4	4.4.2.33
	Body, Firing Pin (Dwg. 9333816) Examination for Defects	25		3.4	4.4.2.34
	Pin Straight (Dwg. 9345321) Examination for Defects	25		3.4	4.4.2.35
	Firing Pin Complete (Dwg. 9345130) Examination for Defects	50		3.4	4.4.2.36
	Cap, Firing Pin (Dwg. 9333817) Examinition for Defects	25		3.4	4.4.2.37

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 8 OF 11	DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231		Listed below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spring, Firing Pin (Dwg. 9333810)				
	Examination for Defects	50		3.4	4.4.2.38
	Spring Embrittlement	25		3.4.4.1	4.5.2
	Spring Set	25		3.4.4.2	4.5.3
	Firing Pin Assembly (Dwg. 9347182)				
	Examination for defects	25		3.4	4.4.2.39
	Ball, Detent (Dwg. 9349223)				
	Examination for Defects	25		3.4	4.4.2.40
	Fuze, Subassembly No. 6 (Dwg. 9347188)				
	Examination for defects	25		3.4	4.4.2.41
	Plate, Barrier (Dwg. 9333809)				
	Examination for defects	25		3.4	4.4.2.42
	Spring, Arming Pin Lock (Dwg. 9345129)				
	Examination for Defects	25		3.4	4.4.2.43

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 9 OF 11		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231				Listed below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc Closing (Dwg. 9333821)				
	Examination for Defects	25		3.4	4.4.2.44
	Lead Cup, Short (Dwg. 9333822)				
	Examination for Defects	50		3.4	4.4.2.45
	Lead Cup Assembly, Short (Dwg. 9333823)				
	Examination for Defects	50		3.4	4.4.2.46
	Lead Output	50		3.4.6	4.5.5
	Lead Cup, Long Dwg. 9333819				
	Examination for Defects	25		3.4	4.4.2.47
	Lead Cup Assembly, Long Dwg. 9333820				
	Examination for Defects	50		3.4	4.4.2.48
	Lead output	50		3.4.6	4.5.5
	Plate Assembly, Barrier (Dwg. 9345166)				
	Examination for Defects	25		3.4	4.4.2.49

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 10 OF 11		DRAWING NUMBER
	M231 All Ways Fuze Assembly				9345168
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for Defects	239		3.4	4.4.2.50
	Jolt	12		3.5.1	4.5.6
	Jumble	12		3.5.2	4.5.7
	Transportation Vibration	20		3.5.3	4.5.8/4.5.15/4.5.16
	Helicopter Vibration	30		3.5.4	4.5.9/4.5.15/4.5.16
	Five Foot Drop	20		3.5.5	4.5.10
	Thermal Shock	30		3.5.6	4.5.11/4.5.15/4.5.16
	Shear Wire Pull Test	115(a)		3.5.7	4.5.12
	Safety Pin Pull Test	115(a)		3.5.8	4.5.13
	Arming Time	115(a)		3.5.9	4.5.14(b)
	Fuzing No-Go	115(a)		3.5.10	4.5.15(b)
	Fuzing Go	115(a)		3.5.11	4.5.16(b)

NOTES: (a) The same fuzes are to be subjected to this test.
(b) Prior to performing tests, condition fuzes for two hours after stabilization at indicated temperatures: -50°F ± 5°F -32 fuzes, 70°F ± 15°F - 51 fuzes, 160°F ± 5°F - 32 fuzes.

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 11 OF 11		DRAWING NUMBER
	M230 All Ways Fuze Assembly				9333825
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	239		3.4	4.4.2.51
	Jolt	12		3.5.1	4.5.6
	Jumble	12		3.5.2	4.5.7
	Transportation Vibration	20		3.5.3	4.5.8/4.5.15/4.5.16
	Helicopter Vibration	30		3.5.4	4.5.9/4.5.15/4.5.16
	Five Foot Drop	20		3.5.5	4.5.10
	Thermal Shock	30		3.5.6	4.5.11/4.5.15/4.5.16
	Shear wire pull test	115(a		3.5.7	4.5.12
	Safety Pin Pull Test	115(a		3.5.8	4.5.13
	Arming Time	115(a		3.5.9	4.5.14(b)
	Fuzing No Go	115(a		3.5.10	4.5.15(b)
	Fuzing Go	115(a		3.5.11	4.5.16(b)

NOTES: (a) The same fuzes are to be subjected to this test.
(b) Prior to performing test, condition fuzes for two hours at indicated temperature: -50°F ± 5°F - 32 fuzes, 70°F ± 15°F - 51 fuzes, 160°F ± 5°F - 32 fuzes.

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of loaded fuzes shall contain:

a. Detonators from not more than two M55 production lots.

b. Short and long lead Assemblies from not more than two production lots.

The quantity in each production lot of fuzes shall not exceed 35,000.

* 4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105. Inspection Level II.

b. Sampling plans for fuze casting. Visual and dimensional inspections of casting before machining shall be in accordance with QQ-A-591d, Section 4, Sampling, Inspection and Test Procedures, Examination paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Slider				9333807
					NEXT HIGHER ASSEMBLY
					9347186/9347219
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Height of triggering mass tab		.40%	3.4	Gage
102	Width of slider		.40%	3.4	Gage
103	Thickness of slider		.40%	3.4	Gage
104	Height of cage hole		.40%	3.4	Gage
105	Location of cage hole from side of slider		.40%	3.4	Gage
106	Diameter of cage hole		.40%	3.4	Gage
107	Location of detonator hole from bottom		.40%	3.4	Gage
108	Depth of detonator hole		.40%	3.4	Gage
109	Distance from slider side to end of rack		.40%	3.4	Gage
110	Location of arming pin hole from front		.40%	3.4	Gage
111	Location of safety wire hole from front		.40%	3.4	Gage
112	Location of safety wire hole from bottom		.40%	3.4	Gage
113	Gear teeth parameters		.40%	3.4	Gage
114	Radius at cage/body interface		.40%	3.4	Gage
115	Perpendicularity of triggering mass tab to slider body		.40%	3.4	Gage
116	Straightness of slider		.40%	3.4	Gage

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Slider				9333807
					NEXT HIGHER ASSEMBLY 9347186/9347219
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of safety pin hole		.65%	3.4	Gage
202	Diameter of detonator hole		.65%	3.4	Gage
203	Thickness of bore rider		.65%	3.4	Gage
204	Width of detonator slot		.65%	3.4	Gage
205	Thickness of cage tab		.65%	3.4	Gage
206	Location of detonator slot		.65%	3.4	Gage
207	Length of bore rider		.65%	3.4	Gage
208	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.2	Body, Fuze				9333824
					NEXT HIGHER ASSEMBLY 9347183
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Porosity in firing pin bore		100%	3.4.3	4.5.1
102	Height of firing pin bore	(a)		3.4.2	Gage
103	Diameter of firing pin bore at slider slot	(a)		3.4.2	Gage
104	Diameter of firing pin cap counterbore	(a)		3.4.2	Gage
105	Width of slider slot	(a)		3.4.2	Gage
106	Depth of slider slot	(a)		3.4.2	Gage
107	Location of front end of slider slot	(a)		3.4.2	Gage
108	Location of side of slider slot	(a)		3.4.2	Gage
109	Depth of bore rider slot	(a)		3.4.2	Gage
110	Depth of cage pocket	(a)		3.4.2	Gage
111	Flange diameter	(a)		3.4.2	Gage
112	Location of pivot hole from side of slider slot	(a)		3.4.2	Gage
113	Counterbore diameter, firing pin cap	(a)		3.4.2	Gage
114	Counterbore diameter, arming pin	(a)		3.4.2	Gage
115	Diameter, center hole	(a)		3.4.2	Gage
116	True position, center hole to arming pin counterbore	(a)		3.4.2	Gage

NOTES:
(a) Reduced sampling based on FED-SPEC-QQ-591E.
Sample size to equal a minimum of 0.1% of a production lot per cavity.

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.2	Body, Fuze		9333824
			NEXT HIGHER ASSEMBLY 9347183

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (cont.)					
117	Radii at bottom of slider slot	(a)		3.4.2	Gage
118	Diameter of anti-twist pin holes		.40%	3.4	Gage
119	Distance between anti-twist pin holes		.40%	3.4	Gage
120	Diameter of detent ball hole		.40%	3.4	Gage
121	Location of detent ball hole from center		.40%	3.4	Gage
122	True position, detent ball hole to pivot thread		.40%	3.4	Gage
123	Location of pivot screw surface from center of firing pin hole		.40%	3.4	Gage
124	Thickness of Flange	(a)		3.4.2	Gage
125	Chamfer on flange	(a)		3.4.2	Gage
126	Location of timer holes from side of slider slot		.40%	3.4	Gage
127	Diameter of safety pin hole		.40%	3.4	Gage
128	Orientation of notches		.40%	3.4	Gage
Minor					
201	Location of side of key from horizontal datum	(a)		3.4.2	Gage
202	Width of Key	(a)		3.4.2	Gage
203	Diameter of hub	(a)		3.4.2	Gage
204	Overall height	(a)		3.4.2	Gage
205	Height of intermediate diameter	(a)		3.4.2	Gage
206	True position, center hole to flange	(a)		3.4.2	Gage

NOTES:
(a) Reduced sampling based on FED-SPEC-QQ-591E.
Sample size to equal a minimum of 0.1% of a production lot per cavity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.2	Body, Fuze				9333824
					NEXT HIGHER ASSEMBLY
					9347183
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor (cont.)					
207	Location of safety wire hole from firing pin bore		.65%	3.4	Gage
208	Location of safety wire hole from bottom of body		.65%	3.4	Gage
209	Outside Diameter	(a)		3.4.2	Gage
210	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:
(a) Reduced sampling based on Fed. Spec. QQ-591E.
Sample size to equal a minimum of 0.1% of a production lot per cavity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Spring, Slider				9333811
					NEXT HIGHER ASSEMBLY
					9347186/9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spring load requirement, max.		.40%	3.4	Gage
102	Spring load requirement, min.		.40%	3.4	Gage
103	Solid height		.40%	3.4	Gage
104	Outside diameter of spring		.40%	3.4	Gage
105	Spring embrittlement		.40%	3.4.4.1	4.5.2
106	Spring set		.40%	3.4.4.2	4.5.3
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Screw, Pivot				9333808
					NEXT HIGHER ASSEMBLY 9347183
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Distance from Pivot hole bottom to underside of screw head		.40%	3.4	Gage
102	Diameter of Pivot hole		.40%	3.4	Gage
Minor					
201	Length of thread to underside of head		.65%	3.4	Gage
202	Diameter of tapered counterbore at bottom		.65%	3.4	Gage
203	Diameter of tapered counterbore at top		.65%	3.4	Gage
204	Poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Fuze Subassembly No. 1				9347183
					NEXT HIGHER ASSEMBLY
					9347184
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Excessive adhesive		100%	3.4	Visual
102	Torque requirement		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Anti-twist Pin		9347220
			NEXT HIGHER ASSEMBLY 9347184

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Hardness		.40%	3.4	Cert.
102	Pin diameter		.40%	3.4	Gage
103	Length of pin		.40%	3.4	Gage
104	Surface finish		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Fuze Subassembly No. 2				9347184
					NEXT HIGHER ASSEMBLY
					9347185
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Final position of anti-twist pin		100%	3.4	Gage
102	Minimum pushout force		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Sleeve, Fuze Body				9345131
					NEXT HIGHER ASSEMBLY
					9347185
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Inside diameter		.40%	3.4	Gage
102	Shear wire hole location		.40%	3.4	Gage
103	shear wire hole diameter		.40%	3.4	Gage
104	Spring lock hole location		.40%	3.4	Gage
105	Perpendicularity of flat cut to datum D		.40%	3.4	Gage
Minor					
201	Flange thickness		.65%	3.4	Gage
202	Step location		.65%	3.4	Gage
203	Key notch location		.65%	3.4	Gage
204	Flange chamfer		.65%	3.4	Gage
205	Outside diameter		.65%	3.4	Gage
206	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Fuze Subassembly No. 3				9347185
					NEXT HIGHER ASSEMBLY 9347186
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Position of staked sleeve		.40%	3.4	Gage
Minor					
201	Depth of stake		.65%	3.4	Gage
202	Diameter of stake		.65%	3.4	Gage
203	Width of stake		.65%	3.4	Gage
204	Pushout requirement		.65%	3.4	Gage
205	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Fuze Subassembly No. 4				9347186
					NEXT HIGHER ASSEMBLY
					9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Trigger				9333812
					NEXT HIGHER ASSEMBLY 9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pivoting shank diameter		.40%	3.4	Gage
102	Spherical Radius		.40%	3.4	Gage
103	Overall length		.40%	3.4	Gage
104	Small tip diameter		.40%	3.4	Gage
105	Length of major O.D.		.40%	3.4	Gage
106	Major body diameter		.40%	3.4	Gage
107	Relieved diameter		.40%	3.4	Gage
108	Length of pivoting shank		.40%	3.4	Gage
109	True position of pivoting diameter to major body diameter		.40%	3.4	Gage
110	True position of tip diameter to major body diameter		.40%	3.4	Gage
Minor					
201	Finish on tip		.65%	3.4	Gage
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9333814
4.4.2.12	Pin, Arming				NEXT HIGHER ASSEMBLY 9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Location of shear wire hole from flange		100%	3.4	Gage
Major					
101	Flange diameter		.40%	3.4	Gage
102	Lock spring groove location from back		.40%	3.4	Gage
103	Diameter of lock spring groove bottom		.40%	3.4	Gage
104	Shank diameter		.40%	3.4	Gage
105	Runout, back of flange to shank		.40%	3.4	Gage
106	Taper		.40%	3.4	Gage
107	Shear wire hole diameter		.40%	3.4	Gage
108	Perpendicularity, shear wire hole to shank		.40%	3.4	Gage
109	Lock spring groove radius		.40%	3.4	Gage
Minor					
201	Flange thickness		.65%	3.4	Gage
202	Distance, center to slot bottom		.65%	3.4	Gage
203	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Wire, Shear		9345128
			NEXT HIGHER ASSEMBLY
			9347187

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire diameter		.40%	3.4	Gage
102	Shear load		.40%	3.4	Gage
Minor					
201	Wire length		.65%	3.4	Gage
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Slider, M55 Detonator Assembly				9347219
					NEXT HIGHER ASSEMBLY 9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Orientation of detonator		100%	3.4	Visual
102	Excessive adhesive		100%	3.4	Visual
103	Detonator deterioration		.40%	3.4.5	4.5.4
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Pin, Safety				9333818
					NEXT HIGHER ASSEMBLY 9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire diameter		.40%	3.4	Gage
Minor					
201	Overall length		.65%	3.4	Gage
202	Length end to center of locking loop		.65%	3.4	Gage
203	Height of locking loop center		.65%	3.4	Gage
204	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRSAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Pinion, Escapement				8833366
					NEXT HIGHER ASSEMBLY
					8833364
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pinion profile		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Gear, Escapement				8833365
					NEXT HIGHER ASSEMBLY
					8833364
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity		.40%	3.4	Gage
102	Gear profile		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Escapement Gear-Pinion Assembly				8833364
					NEXT HIGHER ASSEMBLY 8833353
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gap between gear and pinion		.40%	3.4	Gage
102	Staking improper		.40%	3.4	Visual/Gage
103	Perpendicularity		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Pinion, No. 1				8833360
					NEXT HIGHER ASSEMBLY 8833358
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pinion profile		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Gear, No. 1				8833359
					NEXT HIGHER ASSEMBLY
					8833358
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gear profile		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Visual/Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.21	No. 1 Gear-Pinion Assembly				8833358
					NEXT HIGHER ASSEMBLY 8833353
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Staking improper		.40%	3.4	Gage/Visual
102	Perpendicularity		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Pinion, Number 2				8833363
					NEXT HIGHER ASSEMBLY
					8833361
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pinion profile		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Visual/Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Gear, Number 2				8833362
					NEXT HIGHER ASSEMBLY 8833361
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gear profile		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Visual/Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	No. 2 Gear-Pinion Assembly				8833361
					NEXT HIGHER ASSEMBLY 8833353
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Gap between gear and pinion		.40%	3.4	Gage
102	Staking improper		.40%	3.4	Gage/Visual
103	Perpendicularity		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	Lever				8833369
					NEXT HIGHER ASSEMBLY
					8833367
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		.40%	3.4	Gage
102	Lobe width		.40%	3.4	Gage
103	Pallet hole position from center		.40%	3.4	Gage
104	Pallet hole separation		.40%	3.4	Gage
105	Pallet hole Diameter		.40%	3.4	Gage
106	Center hole diameter		.40%	3.4	Gage
Minor					
201	Lever thickness		.65%	3.4	Gage
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.26	Pallet, Lever		8833368
			NEXT HIGHER ASSEMBLY 8833367

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet diameter		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Gage/Visual
Minor					
201	Pallet length		.65%	3.4	Gage
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.27	Shaft, Lever				8833370
					NEXT HIGHER ASSEMBLY
					8833367
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Surface finish improper		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.28	Lever Assembly				8833367
					NEXT HIGHER ASSEMBLY 8833353
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Staking improper		.40%	3.4	Gage
102	Plating improper		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.29	Plate, Bottom				8833356
					NEXT HIGHER ASSEMBLY 8833353
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of pinion holes		.40%	3.4	Gage
102	Diameter of pinion holes		.40%	3.4	Gage
103	Location of pillar holes		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.30	Plate, Top		8833357
			NEXT HIGHER ASSEMBLY 8833353

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of pinion holes		.40%	3.4	Gage
102	Location of tabs from front flat edge		.40%	3.4	Gage
103	Location of pillar holes		.40%	3.4	Gage
104	Diameter of pinion holes		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.31	Pillar				8833355
					NEXT HIGHER ASSEMBLY
					8833353
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Major diameter length		.40%	3.4	Gage
102	Minor diameter		.40%	3.4	Gage
103	Concentricity of minor diameter		.40%	3.4	Gage
Minor					
201	Major diameter		.65%	3.4	Gage
202	Length of minor diameter		.65%	3.4	Gage
203	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.32	Escapement Gear Assembly				8833353
					NEXT HIGHER ASSEMBLY 9347187
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Axial pull of assembly		.40%	3.4	Gage
102	Component misalignment		.40%	3.4	Gage
103	Stake missing or above flush		.40%	3.4	Visual
104	Maximum height		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.33	Fuze Subassembly No. 5				9347187
					NEXT HIGHER ASSEMBLY 9347188
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety wire missing (a)		100%	3.4	Visual
Major	None defined				
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:
(a) To be verified after arming stroke has been checked.

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9333816
4.4.2.34	Body, Firing Pin				NEXT HIGHER ASSEMBLY 9345130
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large diameter		.40%	3.4	Gage
102	Detent ball angle		.40%	3.4	Gage
103	Concentricity, spring post to large diameter		.40%	3.4	Gage
104	Dowel pin hole diameter		.40%	3.4	Gage
Minor					
201	Stem diameter		.65%	3.4	Gage
202	Concentricity, stem to large diameter		.65%	3.4	Gage
203	Location of assembly wire groove		.65%	3.4	Gage
204	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.35	Pin, Straight				9345321
					NEXT HIGHER ASSEMBLY 9345130
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tip diameter		100%	3.4	Gage
102	Tip edge radius, max.		100%	3.4	Gage
103	Tip angle		100%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.36	Firing Pin Complete	1	1	9345130
				NEXT HIGHER ASSEMBLY 9347182

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tip length		100%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.37	Cap, Firing Pin		9333817
			NEXT HIGHER ASSEMBLY 9347182

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Head diameter		.40%	3.4	Gage
102	Inside diameter		.40%	3.4	Gage
103	Concentricity between head diameter and inside diameter		.40%	3.4	Gage
104	Concentricity between head and shank		.40%	3.4	Gage
105	Runout		.40%	3.4	Gage
Minor					
201	Head thickness		.65%	3.4	Gage
202	Overall length		.65%	3.4	Gage
203	Shank diameter		.65%	3.4	Gage
204	Bottom chamfer		.65%	3.4	Gage
205	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.38	Spring, Firing Pin				9333810
					NEXT HIGHER ASSEMBLY 9347182
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Spring load requirement, max.		.40%	3.4	Gage
102	Spring load requirement, min.		.40%	3.4	Gage
103	Solid height				
104	Outside diameter of spring		.40%	3.4	Gage
105	Spring embrittlement		.40%	3.4.4.1	4.5.2
106	Spring set		.40%	3.4.4.2	4.5.3
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.39	Firing Pin Assembly		9347182
			NEXT HIGHER ASSEMBLY 9347188

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Integrity of firing pin point		100%	3.4	Visual
102	Position of assembly wire		100%	3.4	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

62

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349223
4.4.2.40	Ball, Detent				NEXT HIGHER ASSEMBLY 9347188
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Ball diameter		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.41	Fuze Subassembly No. 6		9347188
			NEXT HIGHER ASSEMBLY: 9333825 or 9345168

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety pin missing		100%	3.4	Visual
Major					
101	Position of detent ball		100%	3.4	Visual
102	Stake diameter		.40%	3.4	Gage
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.42	Plate, Barrier				9333809
					NEXT HIGHER ASSEMBLY
					9345168/9345166
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		.40%	3.4	Gage
102	Thickness		.40%	3.4	Gage
103	Depth of lead counterbore		.40%	3.4	Gage
104	Flatness		.40%	3.4	Gage
Minor					
201	Diameter of lead counterbore		.65%	3.4	Gage
202	True position, Lead hole to O.D.		.65%	3.4	Gage
203	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.43	Spring, Arming Pin Lock				9345129
					NEXT HIGHER ASSEMBLY
					9333825 or 9345168
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire diameter		.40%	3.4	Gage
Minor					
201	Width of opening		.65%	3.4	Gage
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.44	Disc, Closing				9333821
					NEXT HIGHER ASSEMBLY 9333823
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Disc thickness		.40%	3.4	Gage
Minor					
201	Diameter		.65%	3.4	Gage
202	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.45	Lead Cup, Short				9333822
					NEXT HIGHER ASSEMBLY
					9333823
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		.40%	3.4	Gage
102	Material thickness		.40%	3.4	Gage
103	Major diameter of closing disc hole		.40%	3.4	Gage
104	Concentricity between O.D. of cup and closing disc hole		.40%	3.4	Gage
Minor					
201	Overall flange diameter		.65%	3.4	Gage
202	Inside diameter of cup		.65%	3.4	Gage
203	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.46	Lead Cup Assembly, Short				9333823
					NEXT HIGHER ASSEMBLY 9345166
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Explosive height		.40%	3.4	Gage
102	Overall height		.40%	3.4	Gage
103	Explosive density		.40%	3.4	Cert.
104	Lead output	32		3.4.6	4.5.5
Minor					
201	Outside diameter of flange		.65%	3.4	Gage
202	Outside diameter		.65%	3.4	Gage
203	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.47	Lead Cup, Long				9333819
					NEXT HIGHER ASSEMBLY
					9333820
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall Length		.40%	3.4	Gage
102	Material thickness		.40%	3.4	Gage
Minor					
201	Overall flange diameter		.65%	3.4	Gage
202	Inside diameter of cup		.65%	3.4	Gage
203	Outside diameter		.65%	3.4	Gage
204	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.48	Lead Cup Assembly, Long				9333820
					NEXT HIGHER ASSEMBLY 9345166
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall height		.40%	3.4	Gage
102	Explosive density		.40%	3.4	Gage
103	Explosive height		.40%	3.4	Gage
104	Lead output	32		3.4.6	4.5.5
Minor					
201	Outside diameter of flange		.65%	3.4	Gage
202	Outside diameter		.65%	3.4	Gage
203	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.49	Plate Assembly, Barrier (M230)				9345166
					NEXT HIGHER ASSEMBLY 9333825
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Position of short lead		100%	3.4	Visual
102	Damaged lead cup		100%	3.4	Visual
Minor					
201	Evidence of poor workmship		.65%	3.7	Visual

NOTES:

DRDAR-QA Form 160 Jul 77

72

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.50	M231 All Ways Fuze Assembly				9345168
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Jolt	6		3.5.1	4.5.6
2	Jumble	6		3.5.2	4.5.7
3	Transportation Vibration	12		3.5.3	4.5.8/4.5.15/4.5.16
4	Helicopter Vibration	12		3.5.4	4.5.9/4.5.15/4.5.16
5	Five Foot Drop	15		3.5.5	4.5.10
6	Thermal Shock	12		3.5.6	4.5.11/4.5.15/4.5.16
7	Shear Wire Pull Test	32(a)		3.5.7	4.5.12
8	Safety Pin Pull Test	32(a)		3.5.8	4.5.13
Major					
101	Major diameter of fuze		100%	3.4	Gage
102	Height of major diameter of fuze		100%	3.4	Gage
103	Arming time	32(a)		3.5.9	4.5.14(b)
104	Fuzing No-Go	32(a)		3.5.10	4.5.15(b)
105	Fuzing Go	32(a)		3.5.11	4.5.16(b)
Minor					
201	Evidence of poor workmship		.65%	3.7	Visual

NOTES: (a) The same fuzes are to be subjected to these tests.
(b) Prior to performing tests, condition fuzes for two hours at indicated temperature:
$-50^{o}F \pm 5^{o}F$: 16 fuzes, $160^{o}F \pm 5^{o}F$: 16 fuzes

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63446A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.51	M230 All Ways Fuze Assembly		9333825
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Jolt	6		3.5.1	4.5.6
2	Jumble	6		3.5.2	4.5.7
3	Transportation Vibration	12		3.5.3	4.5.8/4.5.15/4.5.16
4	Helicopter Vibration	12		3.5.4	4.5.9/4.5.15/4.5.16
5	Five Foot Drop	15		3.5.5	4.5.10
6	Thermal Shock	12		3.5.6	4.5.11/4.5.15/4.5.16
7	Shear Wire Pull Test	32(a)		3.5.7	4.5.12
8	Safety Pin Pull Test	32(a)		3.5.8	4.5.13
Major					
101	Major diameter of fuze		100%	3.4	Gage
102	Height of major diameter of fuze		100%	3.4	Gage
103	Arming time	32(a)		3.5.9	4.5.14(b)
104	Fuzing No Go	32(a)		3.5.10	4.5.15(b)
105	Fuzing Go	32(a)		3.5.11	4.5.16(b)
Minor					
201	Evidence of poor workmship		.65%	3.7	Visual

NOTES: (a) The same fuzes are to be subjected to these tests.
(b) Prior to performing tests, condition fuzes for two hours at indicated temperature:
$-50^{o}F \pm 5^{o}F$: 16 fuzes, $160^{o}F \pm 5^{o}F$: 16 fuzes

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the Quality Conformance Inspection Tables starting with 4.4.2.1.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6.5 herein. The contractor shall submit all inspection equipment designs to the government for evaluation and approval prior to fabrication of such equipment.

* 4.5 Test methods and procedures.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Porosity check. Prior to performing any secondary machining operations on the casting, the underside and tapered firing pin bore shall be screened by shining a focused beam of light through the bore from the inside of the part. The casting shall be rejected if the diameter of the bore, at any point along its length, shows evidence of a break in the surface, however small, or any other surface irregularity including flow lines.

4.5.2 Spring embrittlement The spring shall be stretched along its longitudinal axis in the case of a coil spring, or bent in the case of a flat spring until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective. All samples subjected to this test will be damaged and shall not be returned to the production lot.

4.5.3 Spring set. The spring shall be measured at its original free length, assembled in a restraining fixture in accordance with 4.4.4 and compressed to a nominal height at .430 inch, placed in an oven and subjected to a temperature of not more than 180°F or less than 175°F for 24 hours. The springs shall be removed from the oven and be cooled to ambient temperature.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:27Z
Check the source to verify that this is the currenversion before use.

After cooling the springs for 24 hours minimum, the springs shall be measured and the measurement shall be compared to the original free length to ascertain compliance with the specified requirements. Any spring that fails the requirement shall be classed defective.

4.5.4 Detonator deterioration. If the total time between original acceptance of any detonator lot and the assembly of that lot into sliders exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-D-14978 immediately before the detonator lot is assembled into the sliders. This test shall be performed by the contractor on detonators selected by the government inspector at the facility assembling the detonators into the sliders.

4.5.5 Lead output. A random sample of Short (P/N 9333823) and Long (P/N 9333820) Lead Cup Assemblies shall be selected. Each Short and Long Lead Cup Assembly shall be assembled to a holding fixture, to be approved by the government, that simulates relative lead position in a fuze. Design of the fixture shall be such as to enable the output end of the Long Lead Cup Assembly to locate within .010 inch of a steel witness plate. The fixture shall also house an M55 detonator positioned above and within .010 inch of the input end of the Short Lead Cup Assembly.

Initiation of the detonator shall result in high order propagation of the Short and Long Lead Assemblies. After the test, the witness plate shall be inspected, using appropriate measuring equipment, for diameter and depth of impression. The diameter shall measure a minimum of .125 inch and the depth shall be a minimum of .015 inch. Any output test that results in an impression less than the requirement shall be considered a failure. A single failure shall reject the lot.

4.5.6 Jolt. A sample of fuzes shall be randomly selected from each production lot and prepared for the jolt test.

The fuzes shall be mounted to special holding fixtures that simulate the fuze seat in a grenade body. The holders shall be designed such as to allow for assembly of the felt take-up pad beneath the fuze assembly compressed to the proper degree. Clamping arrangements, rather than stakes, shall suffice for securing the fuze assembly to the holder providing the fuze is

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

properly seated and remains so for the duration of the test. Fuzes shall be tested without the RAD Assembly Complete (P/N 9345154) attached to the arming pin. The safety pin (P/N 9333818) shall be in place.

After the test the fuzes shall be inspected prior to removel from the holding fixtures and it shall be established that the fuzes are safe to store, handle, use, or dispose of. In order to be considered safe to use, the fuze assembly must meet one of the following requirements:

a. The safety pin must be engaged and the fuze not armed. The shear wire may be cut or firing pin triggered and the fuze shall still be considered safe to handle, store, dispose of use.

b. The shear wire must not be cut and the fuze not armed. The safety pin may be disengaged or the firing pin triggered and the fuze shall still be considered safe to handle, store, dispose of or use.

If any fuze fails to meet either requirement it shall be considered a safety failure. A single safety failure shall reject the lot. Any fuze subjected to this test will be damaged and shall not be returned to the production lot.

4.5.7 Jumble. A sample of fuzes shall randomly be selected from each production lot and subjected to the jumble test. The fuzes shall be tested as bare assemblies without the RAD Assembly Complete (P/N 9345154).

After the test it shall be established that the fuze is safe to handle, store, use, or dispose of. In order to be considered safe, the fuze assembly must meet one of the two requirements listed as criteria for passing the jolt test. Any fuze that fails to meet either requirement shall be considered a safety failure and shall reject the lot. Any fuze subjected to this test will be damaged and shall not be returned to the production lot.

4.5.8 Transportation Vibration. A sample of fuzes shall be randomly selected from each production lot and subjected to the Transportation Vibration Test. Half of the fuzes shall be tested at -65°F and the remaining half at +160°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

In preparation for the test, each fuze shall be secured to holding fixtures that simulate the fuze seat of a grenade body. Design of the fixtures shall be such as to allow for assembly of a felt take-up pad beneath the barrier plate of the fuze compressed to the proper degree. Clamping arrangements shall suffice as a method to join the fuze to the holding fixture provided the fuze is properly seated and remains so for the duration of the test. It shall be established that the clamping apparatus does not alter the resonant frequencies of the fuze assembly.

Prior to the test, the safety pin shall be removed from the fuzes. After the test, prior to removing the fuzes from the holding fixtures, it shall be established visually that the fuzes are safe. To be considered safe the fuze must meet each of the following requirements:

a. The shear wire must not be cut.
b. The slider must be in the full safe postion.
c. The firing pin must not have triggered.

If any fuze fails to meet every requirement listed above, it shall be considered a safety failure. A single safety failure shall reject the lot.

Having established the fuzes to be safe, operability shall be verified. Half of the samples tested at hot and half of those tested at cold temperature shall be subjected to the Fuzing No-Go test. The test shall be performed on the fuzes while still mounted in the vibration fuze holding fixtures.

Any fuze that fails the Fuzing No-Go test shall be considered an operability failure. A single such failure shall reject the lot. Fuzes that pass the Fuzing No-Go test shall be forced to function prior to disassembly from fixtures.

The remaining fuzes tested at hot and the remaining fuzes tested at cold shall be subjected to the fuzing Go test while mounted in the vibration fuze holding fixtures. Fuzes that fail to function shall be considered operability failures. A single such failure shall reject the lot. Fuzes that fail the Fuzing Go Test shall be forced to function prior to disassembly from the holding fixtures.

4.5.9 Helicopter Vibration. A sample of fuzes shall be randomly selected from each production lot and subjected to the sinusoidal cycling described by Curve M in Method 514 of MIL-STD-810C. All fuzes shall be tested at ambient temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

Prior to the test, the fuzes shall be secured to holding fixtures that simulate the fuze seat of a grenade body assembly. Design of the fixtures shall be such as to allow for assembly of a felt take up pad compressed to the proper degree. The methods utilized to attach the fuze to the holding fixture shall not alter the resonant frequencies of the fuze. The safety pin shall be removed from the fuze just before the test is begun.

Operability verification and acceptance criteria shall be identical to those requirements specified in 4.5.8 (Transportation Vibration).

4.5.10 Five foot drop. A random sample of fuzes shall be selected for the drop test as specified in MIL-STD-331. The drops shall be made in the following orientations:

a. Base down
b. 45° base down
b. Horizontal
c. 45° Nose down
d. Nose down

One fifth of the fuzes shall be dropped in each orientation and thereafter it shall be established visually that the fuze continues to be safe to handle and dispose of. For the fuze to be considered safe the following conditions must be met:

a. The safety wire must be engaged and shown to have locked the slider in the full safe position. If the safety wire remains locked in its original position (i.e., the locking loop of the wire engaged with the safety wire post protruding from the fuze body) and it has been established that the detonator is out of line with the firing pin, then the fuze shall be considered safe to handle and dispose of even if the following is evident:

(1) The shear wire is cut
(2) The firing pin has triggered

b. If the safety wire has removed itself from the fuze during the drop test, the fuze shall be considered safe to handle and dispose of only if each of the following conditions are met:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

(1) The shear wire has not been cut,

(2) The slider has not moved from the full safe position (i.e., the bore rider remains completely contained within the fuze body),

(3) The firing pin has not triggered (i.e., the tail of the firing pin continues to protrude outside the fuze beyond the firing pin cap.

Should any fuze tested fail to meet condition a. or b., it shall be considered a safety failure. A single safety failure shall reject the lot. Any fuze subjectd to this test will be damaged and shall not be returned to the production lot.

4.5.11 Thermal Shock. A random sample of fuzes shall be selected from each production lot and subjected to the Thermal Shock Test 113 of MIL-STD-331. The fuzes shall not require special preparations prior to the test.

After completion of the test, it shall be visually determined that neither the shear wire nor safety pin has broken or disengaged. Any such failure shall be considered a safety failure and shall reject the lot. Operability shall be established by subjecting half the fuzes to the Fuzing No-Go test and the remaining half to the Fuzing Go test. Any fuze that fails to pass shall be considered an operability failure. A single operability failure shall reject the lot.

4.5.12 Shear Wire Pull Test. A random sample of fuzes shall be selected from each production lot and subjected to the shear wire pull test. The fuzes shall be mated to a holding fixture that allows an axial load to be applied to the arming pin. A load cell shall be utilized to measure the applied force within 1 (one) pound.

Any fuze of the sample lot that requires less than 16.5 pounds or more than 19.5 pounds to cut the shear wire shall be considered a failure. One failure shall cause rejection of the lot.

After the test, cut shear wires may be replaced and the fuzes returned to the production lot.

4.5.13 Safety Pin Pull Test. A random sample of fuzes shall be selected from each production lot and subjected to the Safety Pin Pull Test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

Prior to the test, the fuze shall be assembled to a nest that will secure it transversely. It shall also be established visually that the shear wire is intact before the pull test is performed.

A load shall be applied to the safety pin and the force required to remove it from the fuze measured.

An effort less than 1 pound or greater than 2 pounds shall be considered a failure. One failure shall reject the lot.

After the test, the safety wire may be reassembled to the fuze and the fuze returned to the production lot.

4.5.14 Arming time. A random sample of fuzes shall be chosen from each production lot. Each fuze shall be assembled to a fixture that enables determination of the time required for the slider to complete its arming stroke which commences at arming pin retraction and ends with uncaging of the trigger. The design of the fixture shall be government approved. Upon release, the slider must move into the armed position in 1.75 $\pm$ .25 seconds.

A fuze that fails to arm within the prescribed time interval shall be considered a reject. A single reject shall reject the lot. Any fuze subjected to this test shall not be returned to the production lot.

4.5.15 Fuzing No-Go. A random sample of fuzes shall be selected from each production lot and subjected to the Fuzing No-Go test.

The fuzes shall be secured with 3 screws to special fuze holding fixtures that simulate the fuze seat in a grenade body. The design of the fixture(s) shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat.

An impulse shall be applied to the fuze in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 180 $\pm$ 20 gravities. Duration of the impulse shall be 1.5 $\pm$ 0.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

After mounting the fuze to the holding fixture but prior to performing the test, the safety pin and shear wire shall be removed. The fuze shall then be allowed to arm by raising the arming pin to the end of its stroke at which time the spring lock will engage.

With the fuze in the armed state, the impulse shall be applied. The Trigger (P/N 9333812) must not release the detent ball and firing pin during the shock. Any fuze that functions during the No-Go test shall be considered a failure. A single failure shall cause the production lot to be rejected.

Fuzes that pass this test must be forced to function before removal from the fuze holding fixture. After passing the Fuzing No-Go test, the fuze must not be re-safed. Partial movement of the trigger with respect to the detent ball will make re-safing the fuze a dangerous operation. After passing this no function test, all fuzes must be subjected to the Fuzing Go Test before removal from the test fixture. Any fuze that fails to function during this test shall be considered a failure. Any single failure shall reject the lot.

WARNING: This test is dangerous and the necessary safety precautions must be taken. Fuze holding fixtures must be shielded. After fuzes are armed prior to this test they must not be handled until forced to function.

4.5.16 Fuzing Go Test. A random sample of fuzes shall be selected from each production lot and subjected to the Fuzing Go test.

The fuzes shall be secured with 3 screws to special Fuze holding fixtures that simulate the fuze seat in a grenade body. The design of the fixture(s) shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat.

An impulse shall be applied to the fuze in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 680 $\pm$ 20 gravities. Duration of the impulse shall be 1.0 $\pm$ 0.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

After mounting the fuze to the holding fixture but prior to performing the test, the safety pin and shear wire shall be removed. The fuze shall then be allowed to arm by raising the arming pin to the end of its stroke at which time the spring lock will engage.

With the fuze in the armed state, the impulse shall be applied. The trigger (P/N 9333812) must release the detent ball and firing pin causing initiation of the detonator. Any fuze that fails to function shall be considered a failure. A single failure shall reject the production lot.

Any fuze that fails the Fuzing Go test must be safely forced to function before removal from the fuze holding fixture. This can be accomplished by increasing the magnitude of the impulse to 1000 gravities minimum. Should the armed fuze still not trigger, a hang-fire has occurred and must be handled accordingly. Provisions must be made that will allow for removal of the failed fuze from the test equipment while maintaining a shielded condition.

WARNING: This test is dangerous and the necessary safety precautions must be taken. Fuze holding fixtures must be shielded. After fuzes have been armed prior to this test, they must not be handled until forced to function.

4.5.17 Data requirements. The contractor shall submit a quality inspection test demonstration and evaluation report in accordance with Data Item Description DI-R-1721 and DI-R-1724 giving the detailed test results.

5. PACKAGING

* 5.1 Packaging and marking. Packaging and marking shall be as specified on drawings 8799112, 9340491, and 9340492, and shall comply with CFR 49, Parts 100-199.

5.2 Special packing requirement. To protect the escapement gear assembly from particulate contamination, each fuze shall be sealed into a 3 inch X 3 inch zip-lock type bag prior to insertion into the loose partition (9340491-2).

* 5.3 Ammunition data cards. Date card information shall be as specified in MIL-STD-1167.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

6. NOTES

* 6.1 Intended use. The M230 and M231 Fuze provide safe arming and fuzing for the grenades used in the 2.75 Inch Rocket Multi-Purpose Submunition (MPSM).

6.2 Ordering Data. See MIL-A-48078.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.
4.5.17	Quality Inspection Defect Report	DI-R-1721
4.5.17	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First Article. When a First Article is required for inspection and approval, see 3.2 and 4.3.

6.5 Submission of inspection equipment for design approvals. See 6.2 of MIL-A-48078. Submit designs as required to Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-QAF-I(D), Dover, NJ 07801.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the current version before use.

6.6 Source control and specification control drawings. Certification of conformance from the supplier certifying that the item complies with the drawing requirements shall be provided to the government representative.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and Contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1330-A188)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RESEARCH & DEVELOPMENT CENTER
ATTN: DRSMC-QA
DOVER, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-63446A	FUZES, GENERAL PURPOSE GRENADE: M230 AND M231 (PRACTICE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.